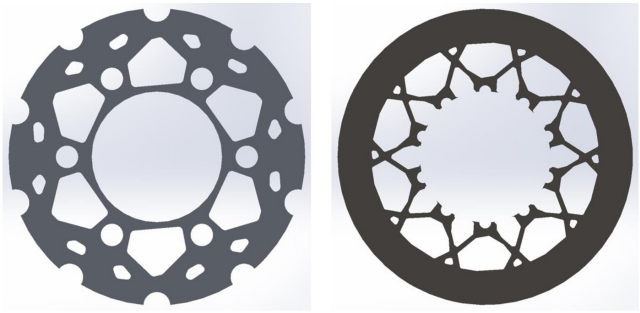
Floating brake disc

Mechanical design, FEA and topology optimisation

SolidWorks / SolidWorks Simulation / Ansys Mechanical | Academic Year 2026/2027



Central carrier and friction ring: final geometries reconstructed in SolidWorks.

-34.3% assembly mass in the design model

OBJECTIVE

Reduce the mass of a motorcycle floating brake disc while preserving functional regions and verifying the structural behaviour of the new geometries.

MY CONTRIBUTION

I created all CAD models, performed all engineering analyses and carried out the optimisation. This was a four-person team project; prototype slicing and printing were carried out within the team.

ENGINEERING APPROACH

- Initial CAD modelling, load definition and static analyses with mesh convergence checks.
- Comparison of parametric and topology optimisation, followed by reconstruction of the final CAD geometries.
- Component and assembly verification, load-sharing sensitivity analysis and fatigue assessment.

RESULTS

In the model, assembly mass decreases from 2,831.6 to 1,861.8 g. Carrier mass falls by 29.1% and friction ring mass by 35.3%. The report identifies finishing of the carrier holes and seats as a requirement for fatigue verification.

SCOPE AND LIMITATIONS

A prototype was 3D printed for form and fit assessment. Braking performance was not experimentally validated; physical testing and thermostructural analysis remain future work.

PROJECT TEAM

Gabriele Petrucci / Catello De Rosa / Francesco Paolucci / Niccolò Maria Fioretti